DATA EXPLORATION AND VISUALIZATION

Units I–V | Part A – 1 Mark Multiple-Choice Questions with Answers

Prepared from the uploaded U25ADG03 study material and question bank. Each question includes all four options and the correct answer.

UNIT I – FOUNDATION OF DATA AND ANALYSIS

1. Which of the following is an example of nominal data?

A) Temperature in Celsius
B) Blood group (A, B, O, AB)
C) Height in cm
D) Age in years
Answer: B) Blood group (A, B, O, AB)

2. The arithmetic mean of 2, 4, 6, 8, 10 is ___.

A) 4
B) 5
C) 6
D) 7
Answer: C) 6

3. The median of an odd-sized ordered dataset is ___.

A) The first element
B) The last element
C) The middle element
D) The average of first and last
Answer: C) The middle element

4. The mode is defined as ___.

A) The average of the data
B) The middle value
C) The most frequently occurring value
D) The range of the data
Answer: C) The most frequently occurring value

5. The formula for population variance σ^2 is ___.

A) $\Sigma(x_i-\mu)^2/N$
B) $\Sigma(x_i-x_j)^2/(n-1)$
C) $\Sigma|x_i-x_j|/n$
D) $\Sigma(x_i-x_j)/n$
Answer: A) $\Sigma(x_i-\mu)^2/N$

6. The Interquartile Range (IQR) is defined as ___.

A) Q3 – Q1
B) Q2 – Q1
C) Q3 – Q2
D) Q4 – Q1
Answer: A) Q3 – Q1

7. A box plot displays ___.

A) Mean and standard deviation
B) Five-number summary
C) Only the mean
D) Only the outliers
Answer: B) Five-number summary

8. A histogram is used to visualize ___.

A) Relationship between two variables
B) Distribution of a single numerical variable
C) Frequencies of categories
D) Time-based trends
Answer: B) Distribution of a single numerical variable

9. Skewness measures ___.

A) Spread of the data
B) Asymmetry of the distribution
C) Central value
D) Number of outliers
Answer: B) Asymmetry of the distribution

10. An IQR outlier is any value ___.

A) Beyond Q3 only
B) Below Q1 only
C) Below Q1−1.5×IQR or above Q3+1.5×IQR
D) Beyond 2 standard deviations
Answer: C) Below Q1−1.5×IQR or above Q3+1.5×IQR

11. When is the median preferred over the mean?

A) When data is symmetric
B) When data is normally distributed
C) When data has extreme outliers or is heavily skewed
D) When all values are equal
Answer: C) When data has extreme outliers or is heavily skewed

12. Adding a constant c to every value changes ___ but not ___.

A) The mean; the standard deviation
B) The median; the variance
C) Both mean and SD equally
D) Neither mean nor SD
Answer: A) The mean; the standard deviation

13. Standard deviation is preferred over variance because ___.

A) It is always larger
B) It is in the same units as the original data
C) It is easier to compute
D) It ignores outliers
Answer: B) It is in the same units as the original data

14. A positively skewed distribution has ___.

A) Mean < Median < Mode
B) Mode < Median < Mean
C) Mean = Median = Mode
D) Mean > Median > Mode
Answer: B) Mode < Median < Mean

15. A larger standard deviation indicates ___.

A) Data clustered near the mean
B) Greater spread and variability
C) Fewer outliers
D) A perfectly normal distribution
Answer: B) Greater spread and variability

16. The z-score measures ___.

A) Frequency
B) How many standard deviations a value is from the mean
C) Rank in the dataset
D) Distance from the median
Answer: B) How many standard deviations a value is from the mean

17. A scatter plot is most useful for ___.

A) Distribution of one variable
B) Relationship between two numerical variables
C) Comparing categories
D) Showing time trends
Answer: B) Relationship between two numerical variables

18. Which visualization compares two group distributions best?

A) Pie chart
B) Single histogram
C) Side-by-side box plots
D) Scatter plot
Answer: C) Side-by-side box plots

19. High kurtosis (leptokurtic) means ___.

A) Flat distribution
B) Heavy tails and a sharp peak
C) Negative skewness
D) No outliers
Answer: B) Heavy tails and a sharp peak

20. The five-number summary is most useful for ___.

A) Calculating the mean
B) Summarizing spread and centre, especially for skewed data
C) Testing hypotheses
D) Measuring correlation
Answer: B) Summarizing spread and centre, especially for skewed data

UNIT II – BIVARIATE AND MULTIVARIATE DATA EXPLORATION

1. Which plot is used to show the frequency distribution of a numerical variable?

A) Scatterplot
B) Histogram
C) Boxplot
D) Heatmap
Answer: B) Histogram

2. What does a boxplot primarily display?

A) Mean and variance
B) Median, quartiles, and outliers
C) Frequency counts
D) Correlation coefficients
Answer: B) Median, quartiles, and outliers

3. Which measure indicates the strength and direction of a linear relationship between two variables?

A) Variance
B) Covariance
C) Correlation coefficient
D) Standard deviation
Answer: C) Correlation coefficient

4. PCA is mainly used for:

A) Feature scaling
B) Dimensionality reduction
C) Outlier detection
D) Frequency analysis
Answer: B) Dimensionality reduction

5. Which visualization displays correlation among multiple variables?

A) Heatmap
B) Scatterplot
C) Histogram
D) Frequency table
Answer: A) Heatmap

6. A correlation coefficient close to 0 implies:

A) Strong positive relationship
B) No linear relationship
C) Strong negative relationship
D) Perfect linear relationship
Answer: B) No linear relationship

7. Why is a heatmap used in multivariate analysis?

A) Display categorical frequencies
B) Visualize correlation among multiple variables
C) Detect outliers only
D) Reduce dimensionality
Answer: B) Visualize correlation among multiple variables

8. When multicollinearity exists in regression:

A) Predictions always improve
B) Coefficients become unstable and unreliable
C) Outliers are removed
D) PCA cannot be applied
Answer: B) Coefficients become unstable and unreliable

9. Why is a violin plot richer than a boxplot?

A) It shows categorical frequencies
B) It combines a boxplot with density distribution
C) It removes outliers
D) It reduces dimensionality
Answer: B) It combines a boxplot with density distribution

10. PCA is a dimensionality reduction technique because:

A) It reduces variables while retaining most variance
B) It removes outliers
C) It scales all features
D) It summarizes categories
Answer: A) It reduces variables while retaining most variance

11. Which transformation is commonly applied to skewed income values?

A) Min-Max scaling
B) Log transformation
C) One-hot encoding
D) Standardization
Answer: B) Log transformation

12. Which visualization compares student scores across departments and detects outliers?

A) Histogram
B) Scatterplot
C) Boxplot
D) Heatmap
Answer: C) Boxplot

13. Which method reduces many features to a few while retaining variance?

A) Covariance
B) PCA
C) Scatter matrix
D) Frequency table
Answer: B) PCA

14. Which visualization compares average salaries across departments?

A) Grouped bar plot
B) Density plot
C) Violin plot
D) Scatter matrix
Answer: A) Grouped bar plot

15. Which measure checks whether two variables move together?

A) Variance
B) Covariance
C) PCA
D) Boxplot
Answer: B) Covariance

16. Why is a pair plot useful?

A) Shows one variable only
B) Compares all variable pairs simultaneously
C) Reduces dimensions
D) Detects categories
Answer: B) Compares all variable pairs simultaneously

17. Multicollinearity causes:

A) More accurate predictions always
B) Automatic outlier removal
C) Unstable and unreliable coefficients
D) PCA failure
Answer: C) Unstable and unreliable coefficients

18. Which measure indicates how two variables vary together?

A) Variance
B) Standard deviation
C) Covariance
D) Mean
Answer: C) Covariance

19. Which transformation brings values into a comparable bounded scale?

A) One-hot encoding
B) Frequency table
C) Min-Max scaling
D) Scatter matrix
Answer: C) Min-Max scaling

20. Which plot compares distributions and detects outliers across departments?

A) Histogram
B) Scatterplot
C) Boxplot
D) Heatmap
Answer: C) Boxplot

UNIT III – TIME SERIES, CATEGORICAL AND TEXTUAL DATA EXPLORATION

1. The four components of a time series are ___.

A) Mean, median, mode, range
B) Trend, seasonality, cyclical, irregular
C) Min, Q1, Q3, max
D) Mean, variance, skewness, kurtosis
Answer: B) Trend, seasonality, cyclical, irregular

2. The trend component represents ___.

A) Short-term random fluctuations
B) Long-term increase or decrease
C) Repeating short-term patterns
D) Business cycle fluctuations
Answer: B) Long-term increase or decrease

3. Lag-k of a time series Y■ refers to ___.

A) Y■ + k
B) Y■■■, the value k periods earlier
C) Y■ × k
D) Y■ / k
Answer: B) Y■■■, the value k periods earlier

4. Autocorrelation measures ___.

A) Correlation between different variables
B) Correlation of a series with its lagged values
C) Trend strength
D) Seasonal amplitude
Answer: B) Correlation of a series with its lagged values

5. A line chart is most suitable for ___.

A) Categorical frequency
B) Text word counts
C) Time series data over time
D) Single-variable distribution
Answer: C) Time series data over time

6. Tokenization means ___.

A) Counting word frequencies
B) Splitting text into individual words or tokens
C) Removing punctuation only
D) Calculating autocorrelation
Answer: B) Splitting text into individual words or tokens

7. Stop words are ___.

A) Most important words
B) Common words carrying little analytical meaning
C) Words appearing once
D) Numeric tokens
Answer: B) Common words carrying little analytical meaning

8. A bar chart visualizes ___.

A) Time trends
B) Frequency or counts of categorical variables
C) Correlation
D) Numerical distribution
Answer: B) Frequency or counts of categorical variables

9. Missing time-series values are commonly handled by ___.

A) Deleting all rows
B) Forward fill, backward fill, or interpolation
C) Setting all to zero
D) Duplicating rows
Answer: B) Forward fill, backward fill, or interpolation

10. Seasonality refers to ___.

A) Long-term direction
B) Irregular noise
C) Patterns repeating at fixed periods
D) Multi-year economic cycles
Answer: C) Patterns repeating at fixed periods

11. Sales peaking every December exhibit ___.

A) Trend
B) Cyclical variation
C) Seasonality
D) Irregular variation
Answer: C) Seasonality

12. A slow multi-year wave in economic data is the ___ component.

A) Trend
B) Seasonal
C) Cyclical
D) Irregular
Answer: C) Cyclical

13. Autocorrelation at lag 1 close to 1 suggests ___.

A) No pattern
B) Strong positive correlation between consecutive observations
C) Stationarity
D) Only lag-12 seasonality
Answer: B) Strong positive correlation between consecutive observations

14. Which visualization shows a categorical distribution?

A) Line chart
B) Scatter plot
C) Bar chart or pie chart
D) Box plot
Answer: C) Bar chart or pie chart

15. Sudden spikes in IoT time-series data may be caused by ___.

A) Annual seasonality
B) Long-term trend
C) Sensor noise or random events
D) Business cycles
Answer: C) Sensor noise or random events

16. For Y={4,6,5,8,7}, the lag-1 series is ___.

A) {4,6,5,8}
B) {4,6,5,8,7}
C) {–,4,6,5,8}
D) {6,5,8,7}
Answer: A) {4,6,5,8}

17. For Y={10,12,14}, mean=12, (Y■–mean)(Y■–mean) equals ___.

A) 0
B) –4
C) 2
D) 120
Answer: A) 0

18. In 'the cat sat on the mat', after removing {the,on}, frequency of 'cat' is ___.

A) 2
B) 0
C) 1
D) 3
Answer: C) 1

19. A repeating 12-period pattern remains after trend removal. This indicates ___.

A) Model is complete
B) Seasonality remains
C) Series is stationary
D) Irregular variation dominates
Answer: B) Seasonality remains

20. For IoT data {22,23,21,22,95,23}, value 95 is best handled by ___.

A) Keeping it
B) Removing entire row
C) Interpolation between neighbours
D) Setting it to zero
Answer: C) Interpolation between neighbours

UNIT IV – DATA VISUALIZATION TOOLS AND TECHNIQUES

1. Too many colours make a chart confusing. Which principle should be applied?

A) Increase chart size
B) Use a consistent colour scheme
C) Add more legends
D) Increase font size only
Answer: B) Use a consistent colour scheme

2. Which Python library is commonly used to create basic plots?

A) TensorFlow
B) Matplotlib
C) OpenCV
D) Django
Answer: B) Matplotlib

3. Which BI tool creates interactive dashboards?

A) Android Studio
B) Power BI
C) Eclipse
D) NetBeans
Answer: B) Power BI

4. Which library is built on Matplotlib for statistical visualization?

A) Plotly
B) Seaborn
C) NumPy
D) SciPy
Answer: B) Seaborn

5. Which chart represents trends over time?

A) Pie chart
B) Line chart
C) Heat map
D) Box plot
Answer: B) Line chart

6. Why use Seaborn for customer age distribution?

A) Database connectivity
B) Better statistical visualization
C) Mobile support
D) Image editing
Answer: B) Better statistical visualization

7. The primary goal of data visualization is to ___.

A) Increase file size
B) Present data clearly for analysis
C) Encrypt data
D) Compress files
Answer: B) Present data clearly for analysis

8. Which tool supports drag-and-drop dashboard development?

A) MySQL
B) Tableau
C) Java
D) VS Code
Answer: B) Tableau

9. Which Google AI assistant can help generate visualization code?

A) BardOS
B) Gemini
C) Photoshop AI
D) TensorRT
Answer: B) Gemini

10. To refresh a D3.js chart dynamically after a dropdown selection, use ___.

A) Static rendering
B) Data update and rebinding
C) HTML tables
D) CSS animation
Answer: B) Data update and rebinding

11. Study hours versus examination scores are best shown using a ___.

A) Pie chart
B) Scatter plot
C) Histogram
D) Bar chart
Answer: B) Scatter plot

12. The missing D3.js SVG container statement is ___.

A) d3.data()
B) d3.select("body").append("svg")
C) d3.transition()
D) d3.scaleLinear()
Answer: B) d3.select("body").append("svg")

13. Calculated measures in Power BI use ___.

A) SQL
B) DAX
C) HTML
D) CSS
Answer: B) DAX

14. A bar chart is preferred over a pie chart for ten categories because it offers ___.

A) Faster loading
B) Easier comparison of multiple categories
C) Lower memory use
D) Better print quality
Answer: B) Easier comparison of multiple categories

15. The Tableau component combining worksheets into one view is ___.

A) Worksheet
B) Dashboard
C) Filter
D) Story point
Answer: B) Dashboard

16. The Power BI feature for interactive filtering is ___.

A) Bookmark
B) Slicer
C) Card
D) Tooltip
Answer: B) Slicer

17. A line chart in a business dashboard is used to ___.

A) Display percentages
B) Show trends over time
C) Compare categories
D) Display distributions
Answer: B) Show trends over time

18. Which feature is available in Tableau but not as a standard Power BI view?

A) Report view
B) Story
C) Data view
D) Model view
Answer: B) Story

19. DAX is primarily a ___.

A) Programming language
B) Formula language
C) Query language
D) Markup language
Answer: B) Formula language

20. A dashboard without axis labels and legends should be improved by ___.

A) Adding more colours
B) Adding titles, labels, and legends
C) Increasing animation
D) Removing gridlines
Answer: B) Adding titles, labels, and legends

UNIT V – STORYTELLING, ETHICS AND COMMUNICATION

1. Which dashboard principle prevents confusion from excessive colours?

A) Interactivity
B) Consistent colour scheme
C) Animation
D) Data aggregation
Answer: B) Consistent colour scheme

2. A dashboard with more than 20 charts violates ___.

A) Interactivity
B) Simplicity
C) Animation
D) Data aggregation
Answer: B) Simplicity

3. For executives, important KPIs should be placed ___.

A) At the bottom
B) Randomly
C) At the top-left or top section
D) On separate pages
Answer: C) At the top-left or top section

4. A better alternative to a pie chart for 20 categories is a ___.

A) Donut chart
B) Bar chart
C) Radar chart
D) Area chart
Answer: B) Bar chart

5. Which action improves dashboard readability?

A) Many font styles
B) Decorative images
C) Consistent colours and spacing
D) 3D charts
Answer: C) Consistent colours and spacing

6. To view data by year and region, add a ___.

A) Animation
B) Filter
C) Background image
D) Watermark
Answer: B) Filter

7. Which visualization supports storytelling for sales growth over time?

A) Scatter plot
B) Pie chart
C) Line chart with annotations
D) Box plot
Answer: C) Line chart with annotations

8. Which is an example of visual bias?

A) Meaningful labels
B) Starting the Y-axis high to exaggerate differences
C) Using legends
D) Displaying sources
Answer: B) Starting the Y-axis high to exaggerate differences

9. Hospital patient data visualization should focus on ___.

A) Attractive colours
B) Decorative icons
C) Clear and accurate information
D) Animated charts
Answer: C) Clear and accurate information

10. Which chart compares sales of five products?

A) Bar chart
B) Line chart
C) Pie chart
D) Area chart
Answer: A) Bar chart

11. A real-time monitoring dashboard should be ___.

A) Static
B) Frequently updated
C) Printed
D) Colourful only
Answer: B) Frequently updated

12. The best way to reduce visual clutter is to ___.

A) Add more colours
B) Remove unnecessary gridlines and labels
C) Add more charts
D) Use multiple legends
Answer: B) Remove unnecessary gridlines and labels

13. A financial dashboard should help users ___.

A) Play animations
B) Make informed decisions quickly
C) Store databases
D) Write reports
Answer: B) Make informed decisions quickly

14. Which visualization is suitable for market share?

A) Line chart
B) Pie chart
C) Histogram
D) Scatter plot
Answer: B) Pie chart

15. Which practice supports ethical visualization?

A) Hide unfavourable data
B) Use distorted scales
C) Show complete and accurate information
D) Change axis values randomly
Answer: C) Show complete and accurate information

16. A dashboard for mobile devices should use ___.

A) Every available chart
B) Responsive layout
C) Fixed screen size
D) Very small fonts
Answer: B) Responsive layout

17. Annotations in storytelling help to ___.

A) Decorate charts
B) Explain important insights
C) Increase dashboard size
D) Add animations
Answer: B) Explain important insights

18. Which principle helps users identify important information first?

A) Data redundancy
B) Visual hierarchy
C) Random placement
D) Multiple fonts
Answer: B) Visual hierarchy

19. Which application commonly uses dashboards for patient monitoring?

A) Agriculture
B) Healthcare
C) Gaming
D) Banking
Answer: B) Healthcare

20. An educational dashboard should primarily include ___.

A) Decorative graphics
B) Student performance indicators
C) Background videos
D) Music
Answer: B) Student performance indicators